PHYS 4041/5041
Fall 2023
Homework 3
Due: In class, Tuesday September 26, 2023

1. Consider the general second-order linear inhomogeneous ODE:

$$y'' + p(x)y' + q(x)y = F(x).$$

A general solution is given by

$$y = y_1 + y_2 + y_p,$$

y_1 and y_2 are linearly independent solutions to the homogeneous equation and y_p is a particular solution to the inhomogeneous equation. Assume here that y_1 and y_2 include the arbitrary constants. Show that one form for y_p is given by

$$y_p = y_2 \int_x \frac{y_1(x)\, F(x)\, dx}{W(x)} - y_1 \int_x \frac{y_2(x)\, F(x)\, dx}{W(x)},$$

where $W(x)$ is the Wronskian:

$$W(x) = y_1 y_2' - y_1' y_2.$$

2. Determine the first two non-zero terms in the Taylor Series expansion of $f(x) = \frac{1}{\sin x} - \frac{1}{\tan x}$ around $x = 0$.

3. Consider the differential equation $(1 - x^2)y'' + xy' - 2y = 0$.
   (a) Find the singular points and classify them (do not worry about what happens as $x \to \infty$).
   (b) Find the Frobenius series around $x = 0$ corresponding to $s = 0$ root of the indicial equation.
   (c) What is its radius of convergence?
   (d) Find the Frobenius series around $x = 0$ corresponding to $s = 1$ root of the indicial equation.

4. Consider the differential equation $x^4 y'' + y = 0$.
   (a) Do you expect the Frobenius Method to work in this case? Why or why not?
   (b) Show that $y_1 = x \sin(1/x)$ and $y_2 = x \cos(1/x)$ are two linearly-independent solutions.
   (c) Is it possible to represent y_1 and/or y_2 as Taylor series exapnsions around $x = 0$?

5. We will find the Green Function $G(r, r')$ satisfying

$$\frac{d^2G}{dr^2} - \frac{l(l+1)}{r^2}G = \delta(r - r'),$$

where $a \leq r \leq b$, $a \leq r' \leq b$, $b > a > 0$, $G(a, r') = G(b, r') = G(r, a) = G(r, b) = 0$, and $l \geq 0$ is a constant.

(a) Find the general solutions to the homogeneous equation by assuming a solution of the form r^m.

(b) Determine $G(r, r')$ from the homogeneous solutions, the boundary conditions, and behavior at $r = r'$.

6. The Hypergeometric differential equation is
$x(x-1)y'' + [(1+a+b)x - c)]y' + aby = 0$, where a, b, and c are constants.

(a) Find the singular points and classify them. Include in your consideration the point $x \to \infty$.

(b) Find the Frobenius series around $x = 0$ corresponding to $s = 0$ root of the indicial equation.

(c) What is its radius of convergence?